SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	**Course Title:**	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	**Assessment:**	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	**Total Marks:**	20 (2 Questions × 10 Marks)
CO5 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)*		
Student Name:		**Reg. No.:**	
Section / Batch:		**Date:**	

Code of Conduct

I __________________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________________________

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
		GRAND TOTAL:	**20 Marks**

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

An automated quality inspection system captures images of metal components. Electrical interference introduces repetitive horizontal stripes across the images, making defect detection unreliable.

(a) Interpret the scenario and identify the type of noise present.
(b) Explain how this degradation affects inspection accuracy.
(c) Recommend an appropriate enhancement technique to remove the interference.
(d) Justify why your selected approach is more suitable than spatial-domain methods.
(e) Explain your answer in a logical sequence.

Scenario 2: Autonomous Vehicle Navigation

A self-driving car captures road images during heavy rain. Although noise is present, the lane markings and road edges must remain sharp for safe navigation.

(a) Interpret the scenario and explain the challenge involved.
(b) Identify the importance of preserving edges while removing noise.
(c) Recommend a suitable filtering technique.
(d) Justify why your selected filter preserves important image details.
(e) Present your explanation clearly and systematically.

Rubrics for Calculation

Criteria	Weig htage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks

Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Faculty In-charge	**Course Coordinator**	**Head of Department**
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

Scenario 1: Manufacturing Quality Inspection

(a) Interpret the Scenario and Identify the Type of Noise

An automated quality inspection system captures images of metal components to identify defects such as cracks, scratches, dents, or holes. However, due to electrical interference in the imaging system, repetitive horizontal stripes appear across the captured images.

Type of Noise:

- **Periodic Noise (Sinusoidal Noise)**
- Appears as regularly spaced horizontal stripes.
- Usually caused by electrical interference, power supply fluctuations, or electromagnetic disturbances.
- It is best analyzed and removed in the **frequency domain**.

(b) Explain How This Degradation Affects Inspection Accuracy

The horizontal stripe pattern negatively affects image quality and reduces the accuracy of the inspection system.

Effects:

- Hides small defects such as cracks and scratches.
- Creates false patterns that may be mistaken as defects.
- Reduces image contrast between the defect and the background.
- Increases false positives (detecting defects that do not exist).
- Increases false negatives (missing actual defects).
- Decreases the overall reliability of automated quality inspection.

(c) Recommend an Appropriate Enhancement Technique

Recommended Technique: Frequency-Domain Filtering using a Notch Reject Filter

Working:

1. Convert the noisy image into the frequency domain using the **Fast Fourier Transform (FFT)**.
2. The periodic stripes appear as bright peaks in the frequency spectrum.
3. Apply a **Notch Reject Filter** to remove only those unwanted frequency components.
4. Perform the **Inverse FFT (IFFT)** to obtain the enhanced image.

(d) Justify Why Your Selected Approach is More Suitable Than Spatial-Domain Methods

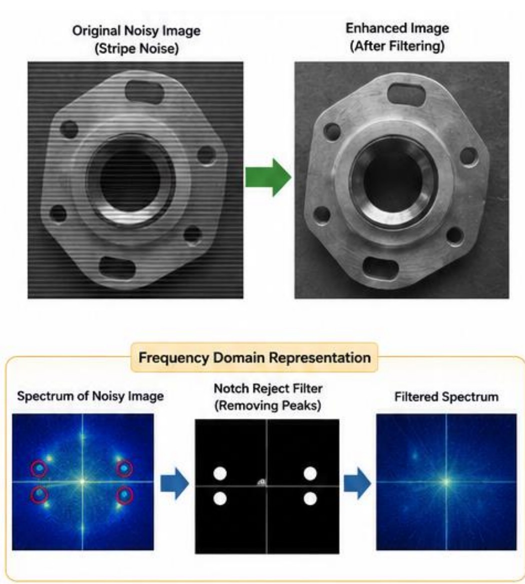
The **Notch Reject Filter** is more effective because periodic noise exists at specific frequencies.

Advantages:

- Removes repetitive stripe noise precisely.
- Preserves important image details such as edges and textures.
- Avoids unnecessary blurring of the image.
- More accurate than Mean, Median, or Gaussian filters for periodic interference.
- Improves defect detection performance.

(e) Logical Sequence of the Solution

Step 1: Capture the image of the metal component.
Step 2: Detect the presence of horizontal periodic noise.
Step 3: Convert the image into the frequency domain using FFT.
Step 4: Apply a Notch Reject Filter to remove stripe frequencies.
Step 5: Perform Inverse FFT to reconstruct the enhanced image.
Step 6: Use the enhanced image for accurate defect detection.



Scenario 2: Autonomous Vehicle Navigation

(a) Interpret the Scenario and Explain the Challenge

A self-driving car captures road images while driving in heavy rain. Rain droplets introduce random noise into the images. At the same time, lane markings, road boundaries, and traffic signs must remain clear for safe navigation.

Challenge:
The system must remove rain-induced noise without blurring important edges and road features.

(b) Identify the Importance of Preserving Edges While Removing Noise

Edges contain essential information for autonomous driving.

Importance of Edge Preservation:

- Helps detect lane markings accurately.
- Preserves road boundaries.
- Enables recognition of traffic signs.
- Improves obstacle detection.
- Supports accurate path planning and vehicle control.
- Prevents accidents caused by incorrect road interpretation.

(c) Recommend a Suitable Filtering Technique

Recommended Technique: Bilateral Filter (Edge-Preserving Filter)

Working:

- Considers both pixel distance and intensity difference.
- Averages only pixels with similar intensity values.
- Smooths noisy regions while preserving edges.
- Removes rain noise without affecting lane markings and road boundaries.

(d) Justify Why Your Selected Filter Preserves Important Image Details

The **Bilateral Filter** preserves edges because it smooths only similar neighboring pixels.

Advantages:

- Removes random rain noise effectively.
- Maintains sharp lane markings and road edges.
- Preserves fine image details.
- Prevents edge blurring.
- Produces clearer images for navigation systems.
- Improves the performance of lane detection and object recognition algorithms.

(e) Logical Sequence of the Solution

Step 1: Capture the road image during heavy rain.
Step 2: Detect the presence of random noise.
Step 3: Apply the Bilateral Filter.
Step 4: Smooth noisy regions while preserving edges.
Step 5: Obtain a clean image with clear lane markings.
Step 6: Use the enhanced image for safe navigation, obstacle detection, and path planning.

Conclusion

- **Scenario 1:** Periodic horizontal stripe noise is best removed using a **Frequency-Domain Notch Reject Filter**, which accurately eliminates interference while preserving important image features.
- **Scenario 2:** Random rain noise is best reduced using a **Bilateral Filter**, which smooths the image while preserving lane markings, road edges, and other critical details required for autonomous vehicle navigation.